L1.2 Limits

Recall: L1.1b Transformations

Name ________________________________ **Solo – 9 minutes**

1. Start from the parent $y = \sqrt{x}$. Write the moves that build

$$g(x) = -\tfrac{1}{2}\sqrt{x+4} + 3$$

in order, each stated as a direction, and mark each one *outside* or *inside*. Then give D and R — interval notation **and** set-builder.

2. Let $f(x) = \dfrac{1}{x-3}$ and $g(x) = \sqrt{x}$. Find $f(g(x))$ and its domain. Write D_{If} and $\mathrm{I}_{f\to\mathrm{O}D}$ separately before you intersect them.

3. Both of these are undefined at $x = 3$, and for different reasons. One sentence each: what does the graph do at $x = 3$?

$$h(x) = \frac{x^2-9}{x-3} \qquad k(x) = \frac{x^2+9}{x-3}$$